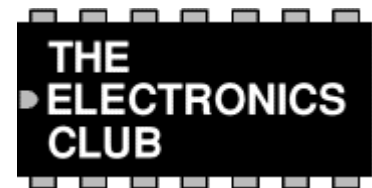


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Stripboard

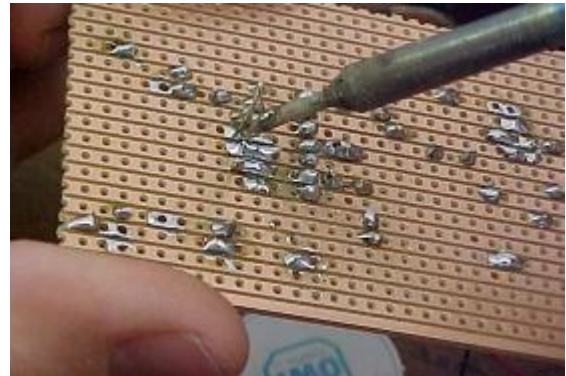


[Placing components](#) | [Cutting tracks](#) | [Planning a layout](#) | [Example plan](#)

Also see: [Breadboard](#) | [PCB](#) | [Types of Circuit Board](#)

Stripboard has parallel strips of copper track on one side. The tracks are 0.1" (2.54mm) apart and there are holes every 0.1" (2.54mm).

Stripboard is used to make up **permanent, soldered circuits**. It is ideal for small circuits with one or two ICs (chips) but with the large number of holes it is very easy to connect a component in the wrong place. For large, complex circuits it is usually best to use a printed circuit board (PCB) if you can buy or make one.



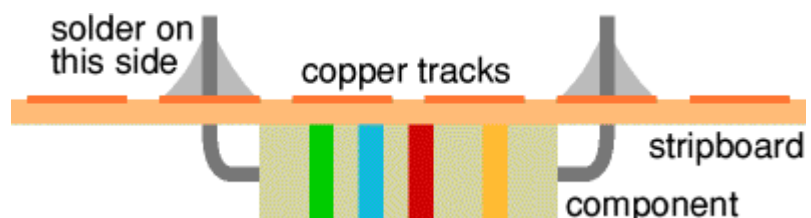
Stripboard circuit (copper tracks side)

Stripboard requires no special preparation other than cutting to size. It can be cut with a junior hacksaw, or simply snap it along the lines of holes by putting it over the edge of a bench or table and pushing hard, but take care because this needs a fairly large force and the edges will be rough. You may need to use a large pair of pliers to nibble away any jagged parts.

Avoid handling stripboard that you are not planning to use immediately because sweat from your hands will corrode the copper tracks and this will make soldering difficult. If the copper looks dull, or you can clearly see finger marks, clean the tracks with fine emery paper, a PCB rubber or a dry kitchen scrub before you start soldering.

Placing components on stripboard

Components are placed on the non-copper side, then the stripboard is turned over to solder the component leads to the copper tracks.



Stripboard layouts are shown from the component side, so the tracks are out of sight under the board. Layouts are normally shown with the tracks running **horizontally** across the diagram.

Placing components on stripboard requires care. The large number of holes means it is very easy to make a mistake! For most small circuits the best method is to very carefully place the chip holder(s) in the correct position and solder in place. Then you can position all the other components relative to the chip holder(s).

Minor position errors left and right will not usually be a problem because the component will still be connected to the correct tracks. However, up and down position errors must be avoided because just one hole too high or too low will connect the component to the wrong track and therefore the wrong part of the circuit.

Some people like to label the holes with letters (up/down) and numbers (across) to give each hole a 'grid reference' but this still requires careful counting of holes.

Cutting stripboard tracks

Most stripboard circuits will need to have some tracks cut to break the connection at that point. This is always necessary under ICs, except for the rare cases where opposite pins must be connected. The tracks are cut with a special track cutter tool or a 3mm drill bit.

Places where the tracks must be broken are usually shown with a cross (X). The cuts are made on the underside (copper side) so extra care is needed to identify the correct hole. It is best to cut the track after soldering because the solder joints will make it easier to identify the correct position.



Track cutter
Photograph © [Rapid Electronics](#)

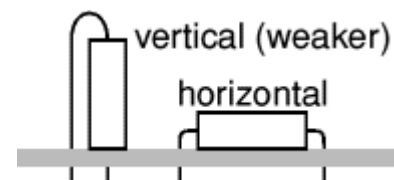
Place the track cutter on the correct hole and twist it to and fro using moderate force. The aim is to break the copper track, not drill a hole through the board! Inspect the cut closely to ensure there is no fine thread of copper left across the break, because even the tiniest piece will conduct.

Planning a stripboard layout

Converting a circuit diagram to a stripboard layout is not straightforward because the arrangement of components is quite different. Concentrate on the **connections** between components, not their positions on the circuit diagram.

Collect all the parts you will be using in the circuit so you can use a piece of stripboard to work out the minimum space they require. For some components (such as chip holders) the space required is fixed, but for others you can increase the space to obtain a better layout. For example most resistors require at least 3 hole-spacings if they are to lie flat on the board, but they can easily span across a greater distance.

If necessary resistors can be mounted vertically between adjacent tracks (0.1" spacing) as shown in the diagram. This arrangement can help to produce a simpler layout but the tracks are more likely to be damaged if the resistor is knocked. If you are designing a stripboard layout for a serious long-term purpose it is best to mount all resistors horizontally.



Plan the layout with a pencil and paper (or on computer if you have suitable software) and check your plan very carefully against the circuit diagram BEFORE you attempt to solder any part of the circuit. The best way to explain the planning process is by example, so there is a step-by-step [example](#) to follow below.

Download a Stripboard Planning Sheet

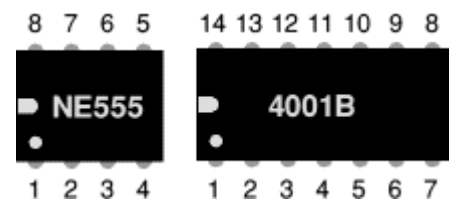
To make planning easier it is best to use paper marked with a 0.1" grid to match the spacing of stripboard holes. You can use graph paper or try our **Stripboard Planning Sheet** which you can [download](#) and print out.

Working 'real size' on a 0.1" grid makes it easy to allow the correct space for components, but you will need to draw very neatly. If you prefer to work at an enlarged scale you can use a piece of stripboard for measuring component sizes in 'number of holes'.

The Planning Sheet is supplied as a PDF file, to view and print it you need an Acrobat Reader which may be downloaded free for [Windows](#), [Mac](#), [RISC OS](#), or [UNIX/Linux](#) computers. If you are not sure which type of computer you have it is probably Windows.

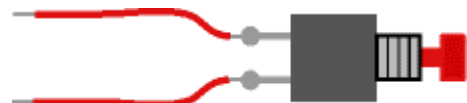
IC pin numbers

IC pins are numbered anti-clockwise around the IC starting near the notch or dot. The diagram shows the numbering for 8-pin and 14-pin ICs, but the principle is the same for all sizes.



Components without suitable leads

Some components such as switches and variable resistors do not have suitable leads of their own so you must solder some on yourself. Use **stranded** plastic-coated wire, single-core wire is not suitable unless the circuit is going to be permanently mounted in a box with no flexing of the wires.



Planning an example stripboard layout

When planning a stripboard layout you must concentrate on the **connections** between components, not their positions on the circuit diagram. The best way to explain the planning process is by example, so the section below explains the process step-by-step for a 555 astable circuit which flashes an LED.

The stripboard tracks are horizontal in all the diagrams.

The circuit diagram

The circuit diagram is the starting point for any stripboard layout, even if you have already built a trial circuit on [breadboard](#).

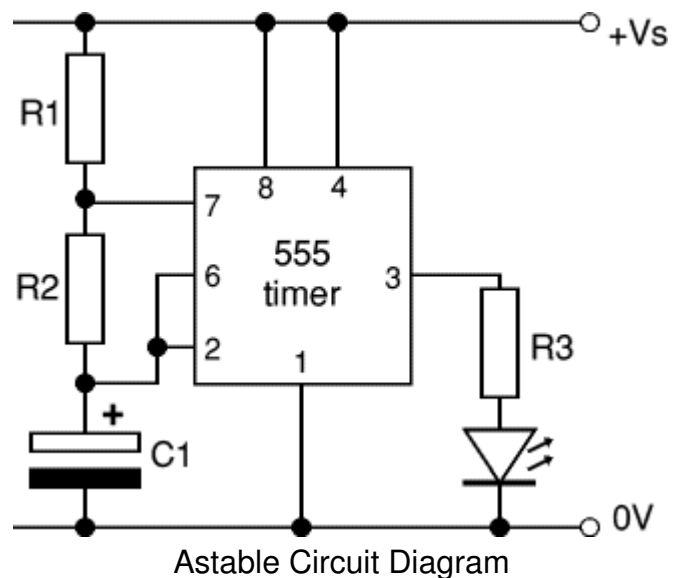
The LED flashes at a rate determined by the resistors R1 and R2 and the capacitor C1. R1 must be at least 1k Ω and both R1 and R2 should not be more than 1M Ω . To select a value for the LED resistor R3 please see the [LEDs](#) page.

LED on time: $T_m = 0.7 \times (R1 + R2) \times C1$

LED off time: $T_s = 0.7 \times R2 \times C1$

$T = T_m + T_s = 0.7 \times (R_1 + 2R_2) \times C_1$
 Frequency (flashes per second), $f = 1/T$
 T_m and T_s are about equal if R_2 is much larger than R_1 .

For further information please see [555 astable](#).



Planning the layout

1. **Place the chip holder near the centre** of your planning sheet with pin 1 at the top left (as in the diagram). You may find it helpful to number the pins.
2. **Mark breaks in each track under the chip holder** with a cross (X). The breaks prevent opposite pins of the chip being connected together. The track beside each pin of the chip is connected to that pin, the diagram shows this for pins 3 and 6.

The four tracks must be broken under the chip holder so they are marked with a cross X

this track is for +Vs

this track is for 0V

blue holes connect to pin 3

green holes connect to pin 6
3. **Mark the power supply tracks +Vs and 0V**, choose tracks which are 2 or 3 spaces above and below the chip holder as shown in the diagram.
4. **Now add the wire links.** Draw a 'blob' (●) at each end of a link. The links are vertical because the stripboard tracks make the horizontal connections. Tinned copper wire (with no insulation) can be used for these links unless there is a risk of them touching other wires (in which case use single core insulated wire). Work round the chip pin-by-pin from pin 1.
 - Draw all the **direct links to the supply tracks** (+Vs and 0V). The diagram shows pin 1 connected to 0V and pins 4 and 8 connected to +Vs.
 - Draw any links required **between pins on the same side** of the chip. There are none in the example, but these links are straightforward to add.
 - **Links to pins on the other side of the chip** require more thought. If the pins

happen to be opposite one another you can erase the track break (X) between them. Otherwise the pins can be linked by connecting both of them to an unused track above or below the chip. The diagram shows pins 2 and 6 linked in this way. Another method is to link them with insulated wire bent around the chip (see the [Flashing LED](#) project for example).

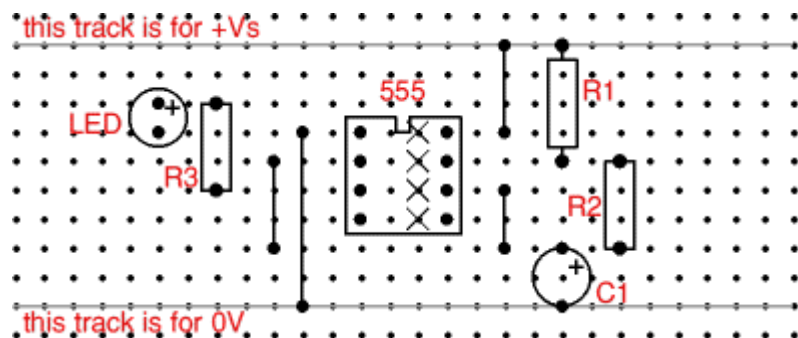
5. **Add components which will be mounted on the stripboard** such

as resistors, capacitors and diodes. Make sure you allow for their size which determines the minimum number of holes, and sometimes the maximum as well.

This is usually the most

difficult stage of planning a layout so expect to change your plan several times!

Remember to **label the components**, otherwise it will become confusing once there are several on the plan.



Connections which do not involve the chip are made using an unused track. For example resistor R3 and the LED are connected by an unused track above the chip.

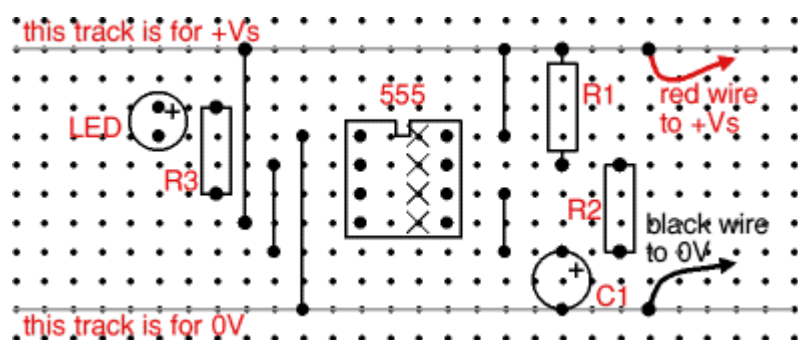
Watch for alternative arrangements using the links you have already made. For example the LED needs to connect to 0V but it is a long stretch to the 0V track. It is easier to connect the LED to the same track as pin 1 of the chip because that track is already connected to 0V by a wire link.

Resistor R2 needs to connect from pin 7 to pin 6 and it could do this directly by mounting it vertically. However, it has been connected from pin 7 to the track used to link pins 2 and 6, the extra space this gives allows R2 to lie horizontally on the board.

6. **Add wires to components which will be off the stripboard** such

as switches. These should normally be on the left and right at the edges of the board. Start by adding the battery clip or power supply leads to the +Vs and 0V tracks.

Connections for the other off-board components are usually easy because you do not need to allow for their size, just draw wires to the correct tracks.



7. **Check your plan very carefully** by checking every connection shown on the circuit diagram. A good way to do this is to work round the chip pin-by-pin. Check all the connections and components connected to pin 1, then move on to pin 2, and so on.

8. **Look for ways to**

.....

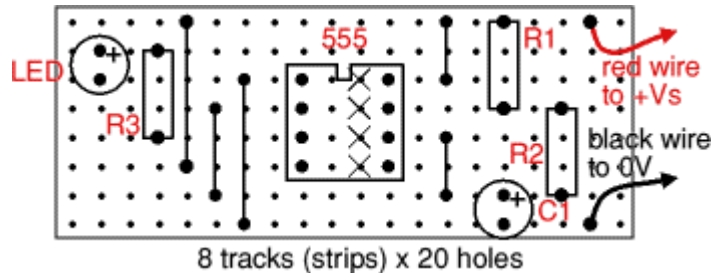
improve your plan. For

example it may be possible to eliminate an unused track by moving a supply track nearer to the chip - but make sure there is still sufficient space for the components. It may also be possible to move links and components closer to the chip horizontally to make the area of board required a little smaller.

Unused tracks above and below the chip have been eliminated in the example. This affected two components, resistor R1 and capacitor C1, but both will still fit in the reduced space. The plan could be compressed a little further by moving components and links closer to the chip horizontally but this has not been done.

9. **Finally, check your plan again and make a neat copy**

fully labelled with all the component references or values. Work out the size of stripboard required. Notice that an extra hole has been allowed on the left and right to avoid soldering at the end of a track. Joints made at the end of a track are likely to break because the small piece of track beyond the last hole easily breaks away from the board.

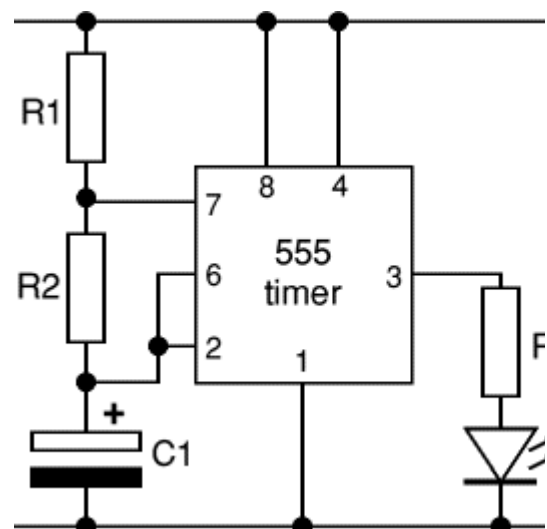
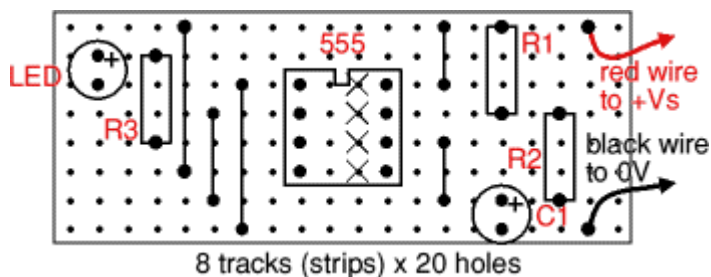


It is tempting to rush straight into soldering the circuit, but do check your plan carefully first. It is much easier to correct errors on the plan than it is to correct them on the soldered board!

This example plan is just one of the many possible layouts for the circuit. The [Flashing LED](#) project uses the same circuit, but the stripboard plan is quite different. In this case the aim was to have the minimum number of wire links.

The completed stripboard layout and the circuit diagram for comparison:

Flashing LED Circuit



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